

Local Projections or VARs: Evidence from Monte-Carlo Simulations

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Motivation

- VARs and Local Projection used for Impulse Response Analysis, think Unemployment and Interest Rate, Oil Prices and Inflation, ...
- finite sample comparison ($\text{VAR}(p)$ vs. $\text{LP}(p)$)
 1. In finite samples, bias-variance puzzle
 2. VARs more efficient, LPs more robust to misspecification
- In Population both estimate the same Impulse Response
- Ludwig identity shows that equal lag comparison is inherently flawed

(BL:) Where does a fixed $\text{LP}(4)$ sit on the VAR complexity frontier and does the conventional equal-lag ranking survive?

(EX:) Does any genuine LP advantage emerge once misspecification is introduced?

Agenda

1. Theory
2. DGP
3. Results
4. Discussion of Results
5. Conclusion

VARs

$$Y_t = \nu + \mathbf{A}Y_{t-1} + U_t \xrightarrow[\text{restr. } K]{A1} y_t = \mu + \sum_{\ell=0}^{\infty} \Psi_{\ell} u_{t-\ell}, \quad \Psi_{\ell} := [\mathbf{A}^{\ell}] \quad (1)$$

Identification: As $\sum_u = \mathbb{E}[u_t u_t'] \neq I_K$, Define $u_t = B\varepsilon_t$ where $\varepsilon_t \sim (0, I_k)$

$$y_{t+h} = \mu + \sum_{\ell=0}^{\infty} \Theta_{\ell} \varepsilon_{t+h-\ell} \quad (2)$$

Suffer from

- LS bias compounds as horizon grows Kilian 1998
- near-unit root behavior
- misspecification through MA dynamics or lag order

LPs

$$y_{t+h} = \alpha_h + \beta_h s_t + \gamma'_h \mathbf{x}_t + \xi_{t+h}^{(h)}, \quad h = 0, 1, \dots, H, \quad (3)$$

Equivalence and Ludwig identity

DGPs

Baseline Results - Complexity Frontier

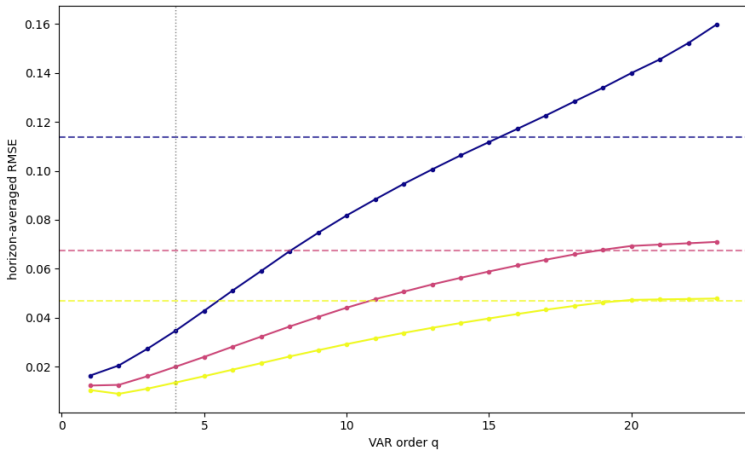


Figure 1:

Complexity Frontier - varying ρ

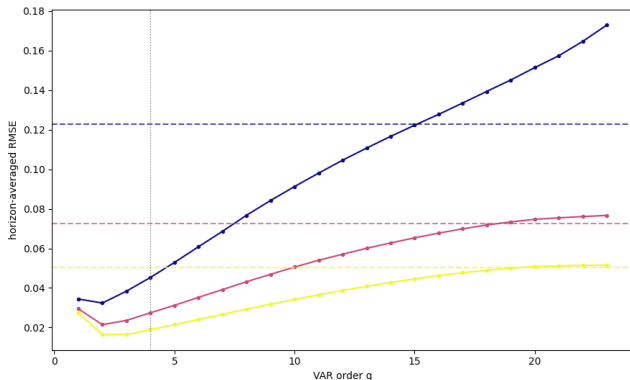


Figure 2: Kausaleffekte von Mentoring auf Sek. I Zuordnung

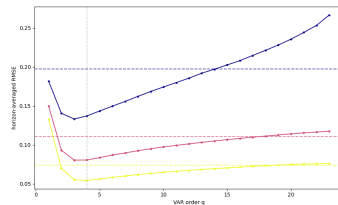


Figure 3: Vergleich Treatment mit Kontrollgruppen

RMSE and Coverage

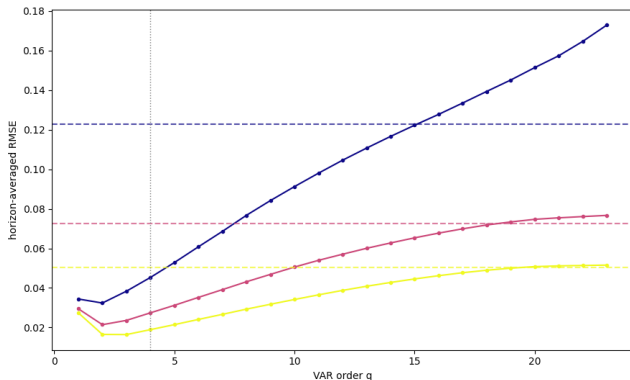


Figure 4: Kausaleffekte von Mentoring auf Sek. I Zuordnung

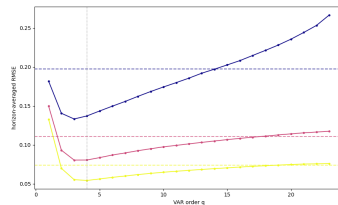


Figure 5: Vergleich Treatment mit Kontrollgruppen

Introducing Misspecification (1)

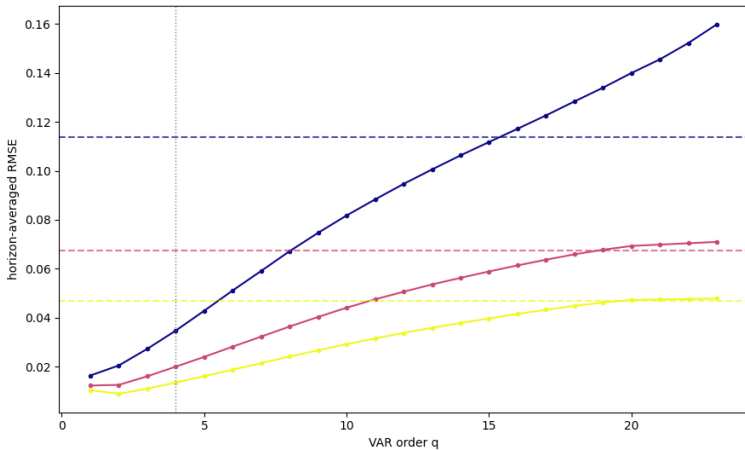


Figure 6:

Introducing Misspecification (2)

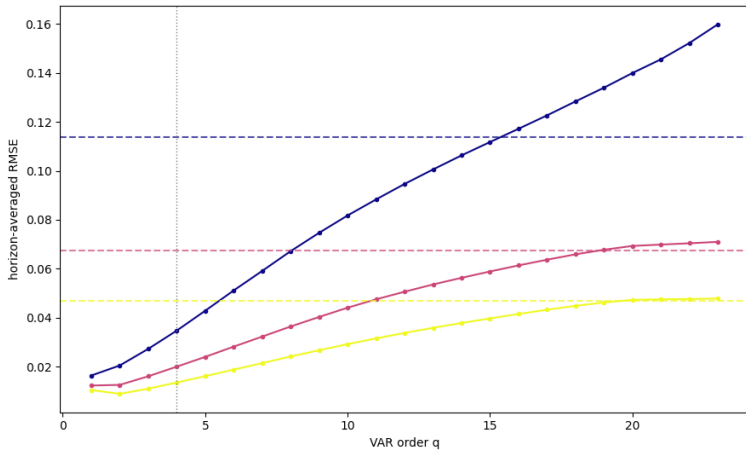


Figure 7:

Discussion of Results

Ludwig identity observable in the data.

Conclusion

- 1.
- 2.
- 3.

Vielen Dank für Ihre Aufmerksamkeit!